

Literature-Implied Answer: Sharp Log-Concave Empirical Covariance Rate

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Status. `literature_implied_answer`. The precise covariance conjecture in arXiv:1005.0816 is proved by the later arXiv:1012.0294. The identification is agent-made: the supporting paper does not explicitly say that it answers Mendelson’s conjecture.

Source conjecture. On printed page 47, near the end of the section “From a bounded diameter to concentration,” Mendelson asks for the optimal empirical covariance rate for independent samples $X_1, \dots, X_N$ from an isotropic log-concave measure on $\mathbb{R}^n$. Equation (6.6), labelled `sampling` in the source, conjectures that for every $N \gtrsim n$, with high probability or in expectation,

$$\sup_{\theta \in S^{n-1}} \left| \frac{1}{N} \sum_{i=1}^N \langle \theta, X_i \rangle^2 - 1 \right| \leq C \sqrt{\frac{n}{N}}.$$

Supporting theorem. Theorem 1 on printed page 2 (source label `raz`) of Adamczak–Litvak–Pajor–Tomczak-Jaegermann proves that if $n \leq N$, the vectors are independent, their one-dimensional marginals have a uniform ψ_1 bound, and their Euclidean norms satisfy the paper’s boundedness condition, then with probability at least $1 - 2e^{-c\sqrt{n}}$,

$$\sup_{\theta \in S^{n-1}} \left| \frac{1}{N} \sum_{i=1}^N (|\langle X_i, \theta \rangle|^2 - \mathbb{E}|\langle X_i, \theta \rangle|^2) \right| \leq C(\psi + K)^2 \sqrt{\frac{n}{N}}.$$

Example 2 immediately after the theorem states that isotropic log-concave vectors satisfy both hypotheses with absolute constants, by Borell’s and Paouris’ estimates. Isotropy turns the expectation in the summand into 1. The result therefore proves the conjectured high-probability estimate, for the slightly larger range $N \geq n$.

Scope and provenance. This identifies an existing proof and is not a new result. It resolves the specific equation (`sampling`); the surrounding informal program of extending other Gaussian results is outside the scope of the packet. The supporting paper describes its result as removing the logarithmic loss and obtaining the Gaussian order for arbitrary N , but it does not explicitly name the source conjecture, so the conservative provenance is `literature_implied_answer`.

References

- [1] S. Mendelson, *Empirical processes with bounded ψ_1 diameter*, Geom. Funct. Anal. 20 (2010), 988–1027; arXiv:1005.0816.
- [2] R. Adamczak, A. E. Litvak, A. Pajor, and N. Tomczak-Jaegermann, *Sharp bounds on the rate of convergence of the empirical covariance matrix*, arXiv:1012.0294.